

# Prabhat Kumar Singh

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## Profile

Full-Stack Software Engineer with 8+ years of experience building and maintaining complex, data-driven web applications across both frontend and backend systems. Strong expertise in modern React, TypeScript, and UI architecture, combined with hands-on backend experience in Java, Spring Boot, and Node.js — including designing and building REST APIs that serve both web and mobile clients. Proven track record of owning features end-to-end in large, evolving codebases, modernizing legacy systems, improving performance, and delivering scalable, maintainable solutions at production scale. Experienced in mentoring engineers, driving technical decisions, and integrating AI-assisted development tools into day-to-day engineering work. Brings a solution-driven mindset with the ability to translate business requirements into effective technical designs, collaborate across engineering and business teams, and consistently deliver high-quality results under pressure.

## Experience

### Evertrue | Netsmartz Infotech | Noida | September 2023 – Present

**Tech stack:** React, JavaScript, Flux-based State Management, Google Maps JS API, Java.

- Led major improvements to Google Maps–based features across core workflows, supporting thousands of user interactions daily and improving performance and usability.
- Designed and delivered advanced filtering capabilities that removed previous technical constraints and enabled new, data-rich user workflows.
- Created a structured testing and modernization framework for legacy UI components that became a reference model adopted by the team.
- Led features end-to-end, owning them from design through release while coordinating and delegating work to peers and junior engineers and remaining accountable for the outcome.
- Took ownership of high-effort, high-complexity initiatives, consistently delivering at ~2× the team's average workload while maintaining quality and stability.
- Designed and built REST APIs powering both the web application and the iOS/Android mobile apps for a mobile-first advancement platform used by 2,000+ nonprofit and education customers, including 1,000+ universities, colleges, and schools.
- Used AI-assisted development tools (GitHub Copilot, Claude, and ChatGPT) daily for coding, debugging, refactoring, and test generation throughout 2025–2026, applying disciplined prompt design, code review, and verification practices to ensure quality and correctness.
- Served as a go-to engineer for complex debugging, diagnosing and resolving critical issues across UI logic, data flows, and third-party integrations.

### Vidsig | Netsmartz Infotech | Noida | July 2023 to September 2023

**Tech stack:** React, TypeScript, Bootstrap, Express JS, Handlebars.

- Owned all frontend development for a global video chat platform, managing both the end-user web application and the internal admin dashboard as a single point of ownership.
- Designed and implemented front-end features supporting customer purchases of one-to-one sessions and expert (talent) onboarding and workflows, translating high-level client requirements into usable interfaces.
- Led UI/UX decisions for both portals, creating layouts, interaction flows, and responsive components in the absence of predefined designs.
- Integrated frontend functionality with backend and mobile APIs, collaborating closely with cross-platform teams to ensure consistent behavior and data flow.

- Maintained and enhanced both the website and admin panel, resolving bugs and delivering stable, high-quality releases through rapid iteration cycles.

#### **Aquant | Netsmartz Infotech | Noida | November 2021 to June 2023**

**Tech stack:** Java, React, HTML Canvas, Video/WebRTC APIs, Heroku

- Worked as a full-stack engineer across frontend (React) and backend (Java), contributing to a video conferencing web application used by customers to connect with technicians for real-time troubleshooting.
- Led development of advanced video call features using HTML Canvas, including annotations, filters, enhanced controls, and recording capabilities, despite limited prior exposure to canvas-based rendering.
- Built and maintained Java backend services and REST APIs supporting both the web application and mobile clients for an enterprise video-triage platform used by global field-service teams across industrial and medtech sectors, collaborating closely with mobile teams to ensure smooth integration and feature parity.
- Owned deployment and environment management for frontend and backend services, maintaining QA, staging, and production environments and handling releases on Heroku.
- Improved the usability and reliability of the video triage experience by iterating client-facing features and stabilizing core workflows across web and mobile platforms.

#### **Appify | Netsmartz Infotech | Noida | August 2021 to November 2021**

**Tech stack:** React, Bootstrap, Redux, Typescript

- Led frontend development across a complex web application maintained in three parallel repositories, each with distinct UI layers and feature sets.
- Delivered high-quality, design-accurate features while accommodating evolving requirements and ensuring delivery within the same or subsequent sprint.
- Acted as a key contributor to cross-project stability, resolving 10+ bugs weekly across multiple client-owned applications.
- Collaborated closely with QA teams to address sprint-based issues, refining functionality, and improving overall product quality.
- Integrated frontend workflows with backend and mobile (iOS and Android) APIs, ensuring consistent behavior and smooth cross-platform communication.

#### **CalSAWS | Deloitte LLC | Bengaluru | December 2019 to May 2021**

**Tech stack:** React (custom framework), Java, Spring Boot (MVC), Oracle → PostgreSQL, REST APIs

- Built and integrated frontend UI components using React.js within a custom framework for a large-scale health benefits web application.
- Worked across frontend and backend layers, developing Java/Spring Boot services and REST APIs consumed by web and mobile clients and integrating them with React-based screens, on a statewide platform consolidating California's 58-county welfare eligibility systems (SAWS programs serving 13M+ residents).
- As part of the core engineering team, onboarded new joiners and delegated and coordinated their work, reviewing output and unblocking issues to keep delivery on track.
- Supported a major modernization effort, migrating legacy systems to Spring Boot, React, and PostgreSQL to improve scalability and maintainability.
- Contributed to end-to-end software delivery, from requirements of analysis and development to testing, debugging, and deployment.
- Assisted in creating technical demos for RFPs and evaluating new project initiatives, supporting business growth and client onboarding.

**Tech stack:** Mainframes, COBOL, Z/OS.

- Maintained and debugged legacy COBOL code on IBM Mainframe (z/OS) systems supporting large-scale healthcare claim processing.
- Investigated failed claim transactions to distinguish between code defects, data inconsistencies, and business-rule violations.
- Collaborated with cross-functional business stakeholders to resolve production issues and ensure accurate, timely claim processing.
- Researched and implemented performance and process improvements, resulting in approximately \$77K in annual cost savings.
- Supported system stability during ongoing modernization efforts from mainframe technologies to modern platforms.

## Education

**Amity University** - Bachelor of Computer Applications, CGPA - 8.3, 2017

### High School

- 12th Grade - 65% (**DAV Kapil Dev Public School**, 2014, CBSE)
- 10th Grade - 90% (**St. Francis School**, 2012, ICSE)

## Skills & Abilities

**Technologies:** JavaScript, TypeScript, React, Next.js, Node.js, Express, REST APIs, HTML5, CSS3, Tailwind CSS, Java, Spring Boot, Python, COBOL (IBM Mainframe, z/OS).

**Databases:** PostgreSQL, Oracle, MySQL, DB2 for z/OS, Firestore, Weaviate (vector database)

**Tools:** Jira, Postman, Git, Docker & Docker Compose, GitHub Actions, IntelliJ IDEA, Eclipse, NetBeans, Visual Studio, Android Studio, SourceTree, Sony Vegas

**Expertise:** Full-stack development, frontend architecture, REST API design & integration, semantic search & vector retrieval, computer vision (MediaPipe, OpenCV), AI-assisted development, mentoring & technical leadership, legacy system modernization, performance optimization, deployment & environment management.

## Personal Projects

Source and written case studies for every project below: [pattotochips.github.io/my-portfolio](https://pattotochips.github.io/my-portfolio).

### **HR Query Engine** — *TypeScript, Express, PostgreSQL, Weaviate, OpenAI, Docker*

Hybrid AI employee search. A GPT pass splits a natural-language query into semantic meaning and hard structured filters; Weaviate handles vector similarity while PostgreSQL enforces the constraints exactly. Includes an embedding cache to bound API cost, CSV bulk ingest, and a documented REST API. ([backend](#), [frontend](#))

### **Face Filter App** — *React 19, MediaPipe, Canvas API, Material UI*

Real-time AI face filters over a live camera feed. Three MediaPipe models (face, hand, pose) share a single render loop, with positional smoothing across frames to eliminate landmark jitter. Includes a particle snow system, a hand-gesture game mode, and a drag-and-drop sequencer so a non-developer can reconfigure the whole experience. ([repo](#))

**WikiTrail** — *Manifest V3 browser extension, D3 v7, vanilla JavaScript*

Records Wikipedia browsing sessions and renders each as an interactive force-directed graph, with dwell time encoded as node size and navigation direction as edges. Per-tab session tracking, annotations, full-text search, and export to PNG, self-contained interactive HTML, or Markdown. One codebase across five browsers. ([repo](#), [live demo](#))

**Expense Splitter** — *React 18, Firebase Auth, Cloud Firestore, Material UI*

Group expense tracker that derives balances from the expense list rather than storing them, then computes the shortest set of transfers to settle a group by greedily matching debtors to creditors. Supports inviting members who have no account yet via pending invites resolved on signup. ([repo](#), [live](#))

**Inkmark** — *Manifest V3 browser extension, vanilla JavaScript, zero dependencies*

Persistent web highlighter with four colors and inline notes. Solves the re-anchoring problem — finding highlighted text again after a page is regenerated — using XPath anchored to the nearest element with a stable unique id, so injected DOM elsewhere on the page cannot invalidate it. ([repo](#))

**OOO Generator** — *Next.js 16, TypeScript, Tailwind CSS, Web Audio API, Jest*

An out-of-office message generator with four tone presets that restructure the message rather than swapping adjectives. Fully deterministic and client-side — no model call, no API key, no data leaving the browser — with a procedurally synthesized 8-bit soundtrack. Unit tested with Jest and React Testing Library. ([repo](#), [live](#))

**Travis Filter** — *Python, OpenCV, MediaPipe, NumPy, PyInstaller*

Real-time desktop video filter compositing a segmented human silhouette over a procedurally animated stripe field. Combines morphological mask refinement with temporal smoothing to stop the silhouette edge flickering frame to frame, with two visual modes and per-mode segmentation tuning. Ships as standalone executable. ([repo](#))

**DeepAR Demo** — *DeepAR Web SDK, JavaScript, Webpack*

Browser AR application demonstrating 19+ real-time face-tracking effects — makeup looks, emotion detection, background replacement, character transforms, and interactive games — behind a carousel selector. Built to evaluate a commercial AR SDK against building the same effects directly from raw landmarks. ([repo](#))

**Birthday Reminder Bot** — *Node.js, discord.js 14, Express, date-fns-tz*

Discord bot that takes reminders as chat commands and announces them in-channel when due. Supports both relative delays and absolute time zone-resolved dates, tracks pending timers per user so they can be listed and cancelled and run a lightweight HTTP health endpoint to stay alive on free-tier hosting. ([repo](#))

**Auto Call Response App (Android), 2017** — *Java, Android Studio*

Android application that automatically sends customizable responses to incoming calls when the device is in silent mode, or the app is active. Tested and validated across Android versions up to Android 4.0 for compatibility and reliable behavior.